

BERGEN, Germany

WMO: 102380

Lat: 52.8153N

Lon: 9.9247E

Elev: 77

StdP: 100.40

Time Zone: 1.00 (EUC)

Period: 97-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.7	-12.2	1.3	-8.8	-9.8	1.6	-6.6	6.6	7.5	6.0	8.1	1.5	80	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	29.7	19.4	27.6	18.5	25.6	17.7	20.3	26.9	19.4	25.8	18.5	24.3	2.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.1	22.2	17.2	12.4	21.5	16.4	11.8	20.7	58.5	26.9	55.5	25.6	52.7	24.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.5	4.1	-14.1	33.8	3.9	2.1	-16.9	35.3	-19.2	36.5	-21.5	37.7	-24.3	39.2		
			-13.5	22.1	4.0	1.1	-16.4	22.8	-18.7	23.4	-20.9	24.0	-23.8	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	1.1	1.6	4.3	8.9	12.9	15.6	17.6	17.6	14.2	9.3	5.0	1.2
	DBStd	7.16	4.49	4.14	3.90	3.87	3.43	3.37	3.26	3.04	3.07	3.60	3.63	4.41
	HDD10.0	1254	275	237	179	66	11	1	0	0	2	57	152	273
	HDD18.3	3433	533	469	435	283	172	95	54	52	129	281	399	531
	CDD10.0	943	0	1	2	33	101	169	237	236	127	35	3	0
	CDD18.3	82	0	0	0	0	3	13	32	29	4	0	0	0
	CDH23.3	1012	0	0	0	6	78	175	384	304	64	1	0	0
	CDH26.7	284	0	0	0	0	13	41	126	96	9	0	0	0
Wind	WSAvg	1.9	2.1	2.1	2.1	1.9	1.8	1.7	1.8	1.6	1.6	1.8	1.8	2.0
Precipitation	PrecAvg	775	75	57	56	45	58	68	78	73	63	61	66	74
	PrecMax	1063	144	125	128	91	140	127	191	140	198	187	129	169
	PrecMin	551	6	15	8	8	21	29	21	28	15	18	3	22
	PrecStd	129	39	28	33	20	29	27	40	32	40	32	32	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	13.0	17.7	24.0	28.4	30.4	33.0	32.5	28.1	20.4	14.8	11.4
		MCWB	10.0	9.5	10.9	14.5	17.5	19.6	19.7	20.1	18.5	16.1	11.8	9.8
	2%	DB	10.4	10.7	15.3	21.5	25.4	27.6	30.2	29.4	25.3	18.2	13.0	9.8
		MCWB	8.8	8.3	9.7	13.3	16.0	18.3	19.6	19.4	17.6	14.5	10.7	8.3
	5%	DB	8.8	9.2	12.7	19.2	23.1	25.2	27.5	26.8	22.5	16.4	11.4	8.4
		MCWB	7.5	7.3	8.9	11.9	14.9	17.3	18.7	19.0	16.4	13.3	9.7	7.2
	10%	DB	7.4	7.6	10.6	16.7	20.8	22.9	25.1	24.3	20.2	14.9	10.2	7.0
		MCWB	6.3	6.0	7.8	11.1	13.8	16.2	17.8	17.7	15.5	12.5	8.7	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.1	12.2	15.5	19.1	20.5	21.7	21.6	19.6	16.6	12.7	10.2
		MCDB	11.3	11.7	15.9	22.2	26.5	27.5	29.4	29.3	25.3	20.0	14.1	10.9
	2%	WB	9.0	8.8	10.6	13.9	17.1	19.4	20.2	20.3	18.1	14.9	11.1	8.6
		MCDB	10.3	10.3	13.4	19.5	23.5	25.7	27.2	26.6	23.6	17.4	12.5	9.6
	5%	WB	7.5	7.5	9.3	12.7	15.6	18.2	19.3	19.3	17.0	13.8	9.9	7.2
		MCDB	8.6	8.9	12.0	17.6	21.3	23.8	26.1	25.1	21.5	15.9	11.2	8.2
	10%	WB	6.4	6.1	8.0	11.5	14.4	17.0	18.4	18.4	16.0	12.7	8.8	6.0
		MCDB	7.4	7.5	10.6	15.9	19.5	21.3	24.1	23.4	19.5	14.5	9.9	6.9
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.8	8.0	10.8	11.4	11.2	10.4	10.6	10.5	8.3	5.3	4.7
		MCDBR	5.7	7.2	12.5	15.5	16.3	15.2	15.5	15.0	14.6	9.9	6.6	5.4
	5% WB	MCWBR	4.7	5.2	7.8	8.2	8.1	7.1	6.4	6.6	7.7	6.5	5.0	4.7
		MCDWR	5.2	5.9	10.3	12.9	14.1	13.4	13.7	12.5	12.5	9.1	6.1	5.2
Clear-Sky Solar Irradiance	taub		0.317	0.336	0.369	0.410	0.409	0.407	0.424	0.417	0.385	0.354	0.334	0.310
		taud	2.369	2.362	2.292	2.218	2.262	2.287	2.251	2.286	2.363	2.435	2.409	2.360
	Ebn at Noon		667	768	811	820	841	846	822	805	787	739	641	601
		Edh at Noon	59	79	103	126	127	125	127	117	96	74	57	50
All-Sky Solar Radiation	RadAvg		0.62	1.31	2.41	3.97	4.91	5.23	4.96	4.16	3.04	1.72	0.76	0.48
	RadStd		0.06	0.18	0.25	0.48	0.55	0.41	0.67	0.32	0.36	0.18	0.08	0.04

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (49 neighbors)	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-142	N/A	N/A	

Nomenclature: See separate page